Annie Prasanna Manoharan

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https://github.com/Annie0159

Education

University of Arkansas Aug 2024 – May 2026
Master of Science in Computer Engineering *Fayetteville, AR*
Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College Aug 2019 – May 2023
Bachelor of Engineering in Electronics and Communication *India*

Technical Skills

Cloud and Tools: Embedded Linux, Verification and validation (V&V), Embedded C, Image quality evaluation, Test automation, Linux, PID control, Image signal processing, Git, SPI/I2C/UART, Multithreading, Xilinx Vivado, Vitis
Machine Learning and AI: Object detection and tracking, Computer vision, PyTorch
Data Engineering:
Languages: C++, C, Python

Professional Experience

Data Engineering Intern Jul 2024 – Present
InsOps Inc. *Chicago, Illinois*

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

Teaching Assistant Sep 2024 – May 2026
University of Arkansas, Dept. of Computer Science and Engineering *Fayetteville, AR*

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant Sep 2024 – Jan 2026
AICV Lab, University of Arkansas *Fayetteville, AR*

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer Jun 2023 – Jun 2024
Soliton Technologies *India*

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements

Projects

Camera Image Quality Analysis Toolkit *C++, Python, OpenCV-style image metrics*

- Build a desktop tool that ingests raw sensor frames and computes objective image quality measures such as noise level, sharpness, and color accuracy to support systematic tuning workflows
- Add automated diagnostics that flag frame-level anomalies like banding, flicker, and exposure drift across captured sequences, with summary reports for cross-team review

FPGA-Based Autonomous Vehicle *Xilinx Arty FPGA, MicroBlaze, Vivado, Vitis, AXI, C*

- Built an autonomous line-following vehicle on a Xilinx Arty FPGA with a custom MicroBlaze-based SoC, integrating IR line sensors, ultrasonic sensors, and motor drivers through AXI IP cores in Vivado
- Wrote embedded firmware in C using Vitis for line tracking, obstacle detection, sensor processing, and motor control
- Implemented Quad-SPI flash boot with an SREC bootloader enabling autonomous startup after power-up